

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL AS BIOLOGY

(9610) BL01

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BLO1 JUNE 2023

There were some excellent responses to questions, with many students using precise biological language appropriately. This was especially evident in question 05.1, where transcription was often described in great detail, and in 07.5, where many students knew the names of specific tests used to classify organisms.

Calculation questions were also generally well answered and most students were able to calculate a percentage change. However, marks were often lost when deciding which data from the stem to use, for example in 07.2 some students selected the incorrect information.

One of the main downfalls of the students was to answer the questions they expected, rather than the ones they received. This was seen in 03.1 and 05.4, where students often listed controlled variables common to enzyme experiments, but these were not always suited to the scenario presented. This was also seen in 05.3, where many students reverted to the idea that different primary structure leads to different tertiary structure, even when this would not answer the question fully.

A significant proportion of students did not seem to understand osmosis correctly, and showed misunderstandings in their answers around the idea that solute rather than just water crosses the membrane/evaporates. A fair few also thought that solute concentration is the same thing as water potential.

Most questions were attempted, and students mostly gave sensible answers to all questions.

QUESTION 01

- 01.1** About half the students gave a correct answer, but there was some confusion amongst the rest. Lots of answers had gas exchange happening at the spiracles, and some ignored the instructions and marked the graph, not the diagram.
- 01.2** Many answered this question correctly and scored full marks. Common answers which scored only 1 mark were '0.07' and less popularly, '50'.
- 01.3** Many students lost marks because they did not finish the comparison fully eg 'insects have spiracles, mammals have lungs' still leaves the possibility open for the mammal to have spiracles, and thus could not be awarded.

The use of spiracle in the singular was not awarded as there are many spiracles in insects.

Lots of students limited their marks by listing the things found on the second marking point (no alveoli/no bronchioles/has tracheoles: alveoli/bronchioles/no tracheoles), and not considering a broader range of structures.

Some students considered chitin and exoskeleton or mentioned features involved in preventing excess water loss, things which were not relevant to the question.

Very few students were awarded the marking point: 'Many tracheae: one trachea'.

- 01.4** Lots of answers were not precise enough eg 'faster gas exchange' with no reference to diffusion was not enough as an alternative to the first marking point (gas exchange direct with muscle/tissue/cells), as overall gas exchange may be faster in mammals.

Only a small proportion of students scored in the disadvantages section, and very few realised that the insect gas exchange system takes up a lot of space.

Answers that scored all the marks available focussed on the gas exchange system and did not dwell exclusively on the circulatory system.

- 01.5** Not many students achieved full marks, and answers were often unclear. Some seemed to suggest causality between percentage change and flutter rate, or just gave lots of data without stating general patterns or fully finishing comparisons. Some students attempted to explain the patterns, which was not needed for the answer.

QUESTION 02

- 02.1** Many students answered this correctly, but if mistakes were made, it tended to be in the carboxyl group. Some students had no idea.
- 02.2** Many answered this correctly and scored 2 marks. Some said peptide bond, some guessed an answer and the most popular mistake was to list one of the products as a protein, which is not the same as a dipeptide.
- 02.3** Most students showed a good understanding of how to interpret the table, got the order correct and scored 2 marks. The most common mistake was to write the order backwards, which could not be awarded a mark because then 'rabbit' was in the wrong relative position.
- 02.4** Few students obtained the mark for the advantages and many just restated the scenario eg 'comparing amino acid sequence can help distinguish between different species.'

Lots of students had the idea of one amino acid sequence being easier to analyse than DNA, but this was not considered mark worthy.

Many students were able to talk about the degenerate code and be awarded the disadvantages mark. Some students missed out on obtaining marks for the third disadvantage as they referenced amino acids being different, and not the idea of different proteins.

QUESTION 03

- 03.1** This question was generally answered badly and very few students scored all 4 marks.

The issues were mostly around imprecise answers, or improvement being suggested that were not specific enough eg stating that the cylinder needs to have the same diameter, but not giving a way of doing that, eg using the same cork borer.

Many students reverted to controlled variables learned from enzyme experiments, but marks were not awarded for 'keep pH the same as the buffer solution needed would affect osmosis. Reference to repeating at different temperatures was also not given marks as it would not help compare the effect of different solution concentrations.

Some students said different intervals, but not more intervals. This was not awarded as there would need to be more intervals between 0.125 and 0.25 mol dm⁻³, in order to obtain a more accurate estimate of the sucrose concentration at 0 percentage change in mass. The idea of adding a control of pure water was also not awarded for the same reason.

Some students said, 'repeat the experiment', but not the idea of it being 3 times or more in total. Even if they scored a mark for the idea of at least 3 repeats, some lost the second mark by implying that lots of repeats actively removes anomalies.

- 03.2** No marks were awarded for suggestion of wearing gloves to prevent scalpel injury, as most latex gloves would not prevent this.

Comments involving protection from burns due to hot water were also ignored, as water only at 30°C, as were comments involving potential corrosive nature of the sucrose solution.

03.3 The vast majority performed this sum correctly and were awarded the full 2 marks.

The most common mistakes were not rounding to the same number of decimal places as the raw data, forgetting to multiply by 100 or getting the equation wrong eg (initial-final)/final.

03.4 Most students mentioned either the different starting mass, or allowing comparisons to be drawn, but only some linked the two ideas together and were awarded the mark.

03.5 Not many students were awarded this mark. Common mistakes involved stating that sucrose/solution entered the melon or mentioned allowing enough time for osmosis to complete/finish. Those that received the mark did so mostly through mentioning the idea of equilibrium being reached, and only a few mentioned constant mass being achieved.

03.6 Most students did not score both marks. Lots did not understand the chemistry of the situation and thought the sucrose/solution would evaporate off. A common mistake was to not give a direction to the change in the water potential, only saying that it would be affected.

A fair number of students were worried about an outside contaminant entering the test tube, which was deemed too unlikely to be awarded a mark.

03.7 Many students gained full marks and showed a good understanding of osmosis, but lots of students lost marks by stating that it is the sucrose solution which moves into the melon, or for not using the words 'osmosis' or 'diffusion'. Some also described the data, rather than explain it, or did not include the idea of water potential in their answers.

03.8 Some students did not know the best way to do this is to draw a graph, and instead discussed how to guess the water potential based on the table. Some proposed additional experiments using different sucrose concentrations. Not many got the third marking point (finding out the water potential of this sucrose concentration), as many thought that sucrose concentration is the same as water potential. Some said water potential of the melon is equal to the water potential outside the melon at 0 percentage change in mass, but that would happen with all of the solutions. The students needed to show that they understood that the sucrose concentration needed to be translated into an actual water potential.

QUESTION 04

04.1 Almost all the students were able to identify the phosphate (group)/ PO_4^- /phosphoric acid. The main way marks were lost was due to the students writing 'ribose' or 'pentose' rather than 'deoxyribose'. The students were expected to be able to identify that the polynucleotide was DNA from the fact that it was double stranded, and hence the more precise answer was needed. Another common mistake was getting answers to Y and X the wrong way round. Some students labelled X as DNA molecule, rather than a nucleotide.

04.2 Most students scored one mark for 'hydrogen bond', with only a few saying phosphodiester, glycosidic, covalent or peptide.

04.3 Most scored well on this, although a surprising number of students seemed unclear on what constituted a monomer in DNA, and there was an array of letters guessed.

04.4 The most common mistake made was to take each species in turn and discuss the proportion of bases, rather than make statements about patterns involving all the species. An exception to this

was when discussing *actinobacteria*. This is because this species was the only prokaryote and so a comparison involving this species would imply a comparison with all the other eukaryotes.

A fair few students identified that complementary bases contained similar percentages but did not score because they did not identify which bases were complementary.

Some students tried to use the numbers to judge the relatedness of different species, which did not focus on patterns and so were not awarded any marks.

- 04.5** Many students scored 2 marks, although fewer identified that the DNA was double stranded than identified the complementary base pairing. Some put the answer to 04.4 here and just described that A and T/ C and G etc have similar percentages. Some mentioned complementary, but did not have the idea of pairing, so could not be awarded the mark. Because the question was asking about the structure, there needed to be the idea of the complementary bases pairing or joining together.
- 04.6** The majority of students scored one mark for an answer of 130, and some others managed to get a mark for referencing 26%. A small proportion of students managed to score 2 marks with an answer of 260, as they realised that 500 base pairs means a starting point of 1000 bases.

QUESTION 05

- 05.1** This question was very well answered by some students. Common mistakes that lost marks included saying that the DNA strands 'unwind' or that the mRNA is made by DNA polymerase.

Some students misread the question and spend half of their time describing translation, which sometimes limited the marks available to them.

- 05.2** The overwhelming majority got this correct, although some forgot that U is only found in RNA and T in DNA or switched them around. Some misunderstood the question and wrote the DNA strand as a complementary version of the mRNA strand.

- 05.3** Few students were awarded all three marks here. Some had achieved the first marking point for the idea that the mRNA (transcribed) from the inserted gene binds to mRNA (transcribed) from the aromatase gene, but then did not go into enough detail for the next stages in translation to get both available marks.

Many students went down the path of the inserted DNA being transcribed and translated into a different protein. This was not awarded, as the aromatase would still be produced from the original DNA. A few stated the new protein produced would interfere with aromatase activity, but this was not awarded as there is no evidence in the stem for this.

Some answers showed up students' misunderstanding of protein synthesis, as they suggested that mRNA from the inserted gene could outcompete the aromatase mRNA at the ribosome, or not have any anticodons available to it.

A few students suggested that the inserted DNA could bind to the aromatase gene and prevent the mRNA from binding. This was not accepted as a plausible suggestion, as it would prevent the DNA strands from joining back together afterwards.

- 05.4** Some students did not score because they mentioned keeping the lake the same, and the stem states there is only one lake. As such, any mention of diet and lifestyle also did not score, as this would not have been feasible to control in wild carp. There were also no marks available for mentioning aromatase enzyme concentration as this is the independent variable.

Some students lost marks by focussing on just one idea eg size and type and freshness of the tissue would only score one mark as they are all on the same marking point.

Students that mentioned controlling substrate concentration mostly did not score the mark because they needed to mention excess substrate, as otherwise enzyme activity could not be fully compared.

Some students thought that the sample sizes needed to be equal for them to be compared, which would not be true in this scenario.

- 05.5** Lots of students stated that carp cannot reproduce which did not score, as some carp would still be able to reproduce.

Those that stated that the carp would not be adapted to the environment also did not score, due to this answer not being specific enough unless linked with a suitable example.

Some students seemed to bring A2 knowledge of genetically modified organisms to the answer, and this was awarded if it was specific enough.

QUESTION 06

- 06.1** Most students scored by stating 'centromere,' although a fair few said 'centrosome'.

The main mistake students made was to pluralise or singularise inappropriately eg when labelling C many said chromosomes, which could not be awarded.

- 06.2** Most students scored marks by saying 'meiosis', but some detailed the phase instead. This was awarded if in conjunction with meiosis eg 'meiosis prophase 1'.

- 06.3** This question was generally well answered with many students giving clear accounts and using correct terminology eg homologous chromosomes. Some students showed knowledge of the word 'chiasma' even though this is not on the specification, and this was awarded when seen.

A small number of students lost marks by mentioning genes not alleles. A fair few did not make it clear that the crossing over was occurring between homologous chromosomes and so couldn't be awarded the second mark.

Some students misinterpreted the question and described metaphase/anaphase 1 rather than focussing on the crossing over.

QUESTION 07

- 07.1** Most students scored 1 or 2 marks for this question. The most frequently made mistakes were found in the 'contains 2 membranes' row. Some students forgot that nuclei have a double membrane, whilst some indicated that prokaryotes do contain this feature.

- 07.2** The majority of students forgot to divide by 2 (the nucleus has 2 copies of each gene) and obtained only one mark for the answer 27 500.

A reasonable proportion of students included the 37 mitochondrial and 2000 nuclear genes in their answer and ended up with the answer 50.875.

Some divided the 27500 by 2000 rather than 2.

- 07.3** Many students got the idea of more copies of the gene, but few linked it to the idea that this would make the gene easier to detect.

There were lots of answers around mDNA being easier to extract because it's not in the nucleus, but this was not awarded as it isn't necessarily true.

Some students gave answers that would decrease comparability, not increase it. This included the idea that fewer genes would make it easier to compare, or that mDNA changes less frequently than nuclear DNA, which would make it more reliable.

A small proportion of students thought that using the maternal DNA to trace their mother would be the surest route to the criminal.

- 07.4** Many students correctly discussed meiosis and fertilisation and were awarded 2 marks. A fair number talked about the possible reasons for a larger mutation rate in DNA eg longer length, histones etc. This was not awarded a mark, as in reality mDNA has a higher mutation rate.

Some students were not awarded the second marking point (fertilisation does not occur), because they did not make it clear that the DNA in the mitochondria is ONLY inherited from the mother.

- 07.5** Most students were able to list 2 methods, the most popular (rather impressively) being 'DNA hybridisation' and 'immunological comparison'. Some students wrote answers involving DNA sequencing or protein sequencing, which was not awarded any marks, as these were already mentioned in the stem. 'Frequency of alleles' was also not awarded due to this being too similar to amino acid sequencing.

Some students missed out on a mark because they did not specify the type of RNA in their sequencing.

Some students mentioned 'anatomy' and 'morphology' as 2 separate points, but these are both found on the same marking point, so could only be awarded one mark.

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